

IDS GROUP ASSIGNMENT

Measuring Progress Towards the UN Sustainable Development Goal 8: Decent Work and Economic Growth

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1. Abstract

In 2015, The United Nations adopted the 2030 Agenda for Sustainable Development and its 17 Sustainable Development Goals (SDGs), aiming to end poverty, protect the planet and tackle inequalities. In this report, we have assessed the progress of six continents – Africa, Asia, Europe, North America, South America, and Oceania – towards meeting SDG 8: Decent Work and Economic Growth. In particular, we performed exploratory data analysis to assess how these continents are faring with respect to Target 8.1 and 8.6.

2.1 Introduction

Sustained economic growth is a key factor in improving standards of living by enabling long-term development. As a result, the UN introduced Target 8.1 under Sustainable Development Goal 8, challenging countries to maintain GDP per capita growth, in accordance with national circumstances, while Least Developed Countries (LDCs) meant to aim for at least 7% growth per annum.

Growth in GDP per capita is widely recognised as an indicator of economic performance, reflecting developments in population welfare as well as changes in productive potential. Our report investigates the extent to which countries across different continents reached growth benchmarks in our chosen time frame of 2015-2021. Consequently, both overall continental results and specific outcomes for LDCs are examined, clearly separating the short-term disruption brought on by the COVID-19 pandemic from structural patterns of long-term development.

2.2 Methodology

2.2.1 Research Design

This study can be separated into two parts. The first part evaluates whether countries in all six continents

managed to “Sustain per capita economic growth in accordance with national circumstances”. This analysis examined the GDP per capita percentage growth rates from 2015 to 2021 for the six continents and compared how each continent performed relative to other continents and the world average within their respective income group, as classified by the World Bank. We chose the time frame 2015 to 2021 because SDG 8 was established in 2015, so we decided that it made the most sense to observe how continents were faring after the goal was established. The second part of this study evaluates whether the least developed countries across the six continents managed to reach the target of a sustained minimum of 7% GDP per capita growth per annum in the same time frame.

2.2.2 Extra CSV File

We chose to use World Bank data with information about the thresholds for income groups using the World Bank’s classifications (as discussed in 2.2.3 below). This is a reliable source of data due to its rigorous adherence to international standards, guided by professional standards in the collection, compilation and dissemination of data.

2.2.3 Measure of “National Circumstances” and Threshold for “Least Developed Countries”

To evaluate how countries in each continent performed in accordance with national circumstances, we divided the countries into four income groups, following the World Bank’s classifications for each income group. The classification is as follows (in GDP per capita PPP constant 2017 international dollars):

Low-income countries (L): $\leq 1,025$

Lower middle-income countries (LM): 1,026 - 4,035

Upper middle-income countries (UM): 4,036 - 12,475

High income countries (H): > 12,475

These classification values are for 2015, with the boundaries for each group slightly changing over the years.

For the “least developed countries” we chose all the countries that fall into the low-income group (L), as classified by the World Bank.

2.2.3 Analytical Methods

Analysis for the first part proceeded in the following stages. Firstly, we calculated the GDP per capita percentage growth rates per annum for all countries for each year from 2015 to 2021, and then we calculated the average growth rates for each country over the entire time range. Secondly, we divided the countries into their income groups, as classified above, and grouped them by continent. Thirdly, we calculated the average growth of all the countries together in a specific income group and continent. Then, we compared how each continent within each income group performed in terms of their growth rates relative to the rest of the continents in that income group. We then plotted bar graphs of the average growth of each continent by income group. Finally, we plotted a boxplot of the average growth from 2015 to 2019 for all countries and grouped them by continent (as in Figure 2.3.2) to visualise the average growth from 2015 to 2021 more easily. When producing the boxplot, we decided to only include the years 2015 to 2019, so that we could analyse average growth without the effects of COVID-19. This was so we could distinguish between the progress of each country independent of the recession which negatively impacted all continents.

For the second part, analysis proceeded as follows. We calculated the GDP per capita percentage growth rates per annum for each low-income country from 2015 to 2021 and then calculated the average growth rates for each low-income country over the entire

time range. Moreover, we calculated how many low-income countries there are in each continent and how many of those reached the target of at least 7% average growth per annum over the specified years. Finally, we created a data frame/table showing all the above results for the low-income countries.

2.3 Results

2.3.1 Investigating Sustained Per Capita Economic Growth

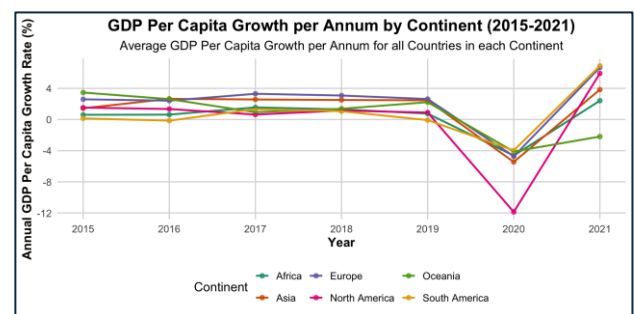


Figure 2.3.1.1 GDP Per Capita Growth per Annum by Continent (2015-2021)

When investigating how each continent has been faring with respect to Target 8.1 under Sustainable Development Goal 8, we chose to analyse the average annual GDP per capita growth of the countries within each continent (excluding Antarctica). Our results can be seen in **Figure 2.3.1**. Until 2019, all continents’ per capita economic growths were positive. Africa clearly shows wide variation and low median growth, while Europe and North America had evident stability, but low overall growth. Asia, however, had the highest average GDP per capita growth, as can be seen in **Figures 2.3.1.1** and **2.3.1.2**.

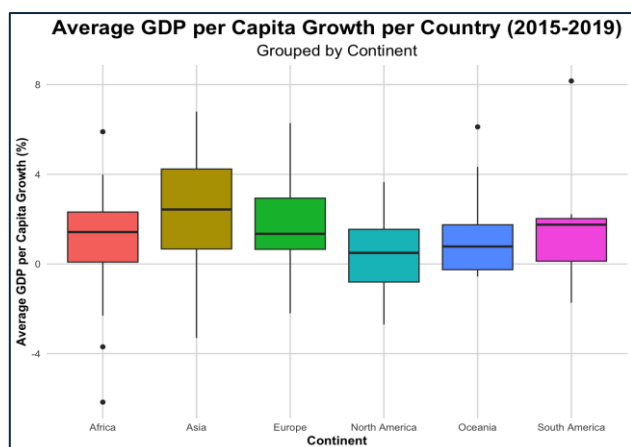


Figure 2.3.1.2 Average GDP per Capita per Country (2015-2019)

hampered Australia's recovery from COVID-19. In the heat maps below, the GDP growth of all continents in 2020 and 2021 is visualised clearly, with green representing growth and red representing loss. The heat maps for 2015 to 2019 can be found in **Appendix A**.

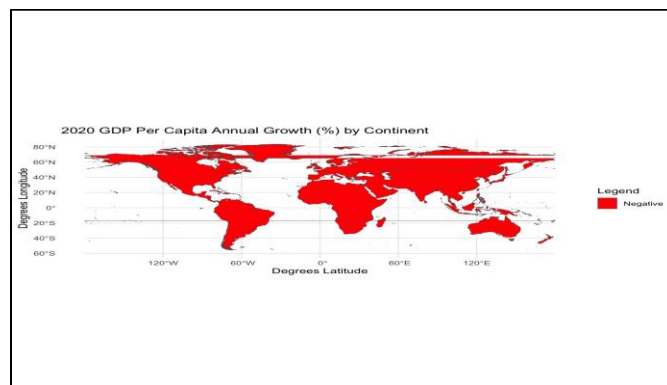


Figure 2.3.1.3 Heat map of GDP per capita growth for each continent from 2019 to 2020

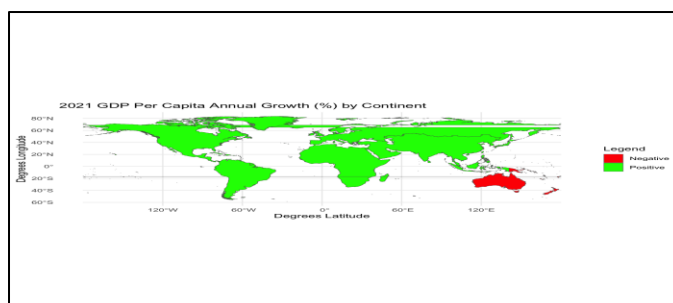


Figure 2.3.1.4 Heat map of GDP per capita growth for each continent from 2020 to 2021

In fact, the continent accounted for more than 57% of the world GDP growth from 2015 to 2021. In particular, Asia had a higher growth rate than the other continents from 2015 to 2016. This reflects Asia's diverse trade relationships, advanced technology, and status as the world's industrial hub. (Seong *et al.*, 2023) By contrast, from **Figure 2.3.1.2**, we can see that North America had the lowest average GDP per capita growth from 2015 to 2019. However, by observing the tails of the boxplots, we see that the spread of average GDP per capita growth was larger for Asia than all other continents, with South America and Oceania having the most consistent spreads. This supports the commentary of Seong *et al.*, which highlights that there is great disparity between Asian countries. This means that more integrated countries such as China greatly influence the average growth, so it is difficult to determine whether the performance is uniform across all Asian countries. While COVID-19 caused universal contraction (with global goods trade falling by 18.5% (Lim *et al.*, 2021)), from **2.3.1.1** we can see that North America had the greatest drop in per capita GDP. However, it also had the greatest rebound compared to other continents in 2021. By contrast, Oceania ended up with the lowest GDP per capita. This is due to Oceania being hit the hardest by the collapse of the tourism and higher education industries in Australia due to travel restrictions. The population growth overtook GDP growth, which also

2.3.2 National Circumstances

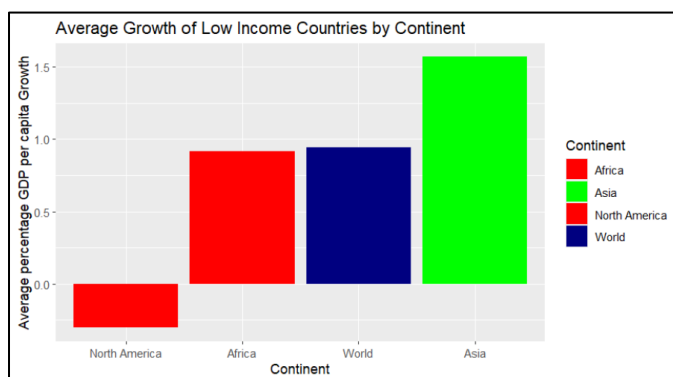


Figure 2.3.2.1

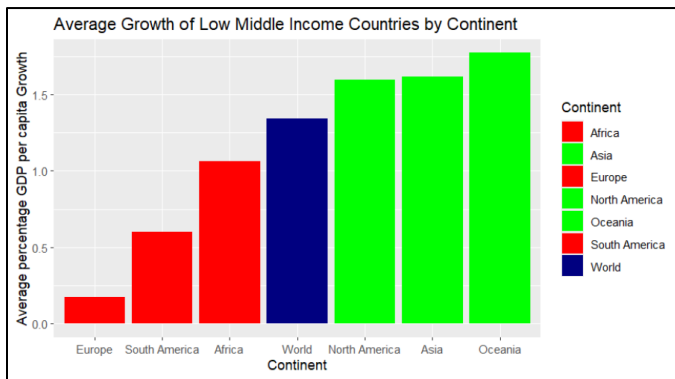


Figure 2.3.2.2

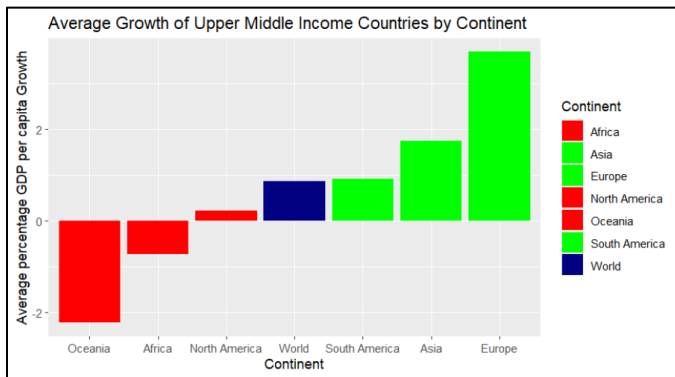


Figure 2.3.2.3

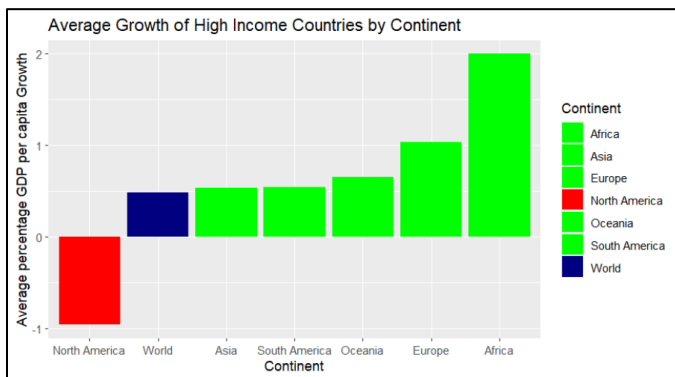


Figure 2.3.2.4

continents had growth, with North America, Asia and Oceania achieving above world average and Europe, South America and Africa achieving below world average. The upper-middle-income group had the greatest inconsistency between continents. South America, Asia and Europe achieving growth above world average, North America achieving negligible growth below world average, and Oceania and Africa incurring a loss. Finally, the upper-income group had the greatest number of continents achieving growth above world average, with only North America incurring a loss. In conclusion, Asia had average GDP growth consistently above the world average across all income groups, showing sustained per capita economic growth across all income groups. It is notable that Africa had the highest average GDP growth in high income countries but was below the world average in other income groups, suggesting high income inequality between African countries. Low-middle-income groups show growth in all continents. However, upper-middle-income groups show the most variation across continents. Overall, the bar graphs taking income group classification into account helps explain cross-continental differences that are not observed from the other graphs.

2.3.3 Investigating GDP Growth per Annum in Least Developed Countries (LDCs)

Regarding the “national circumstances” aspect of Target 1, we chose to compare continents according to their income groups, as classified by the United Nations. In the low-income group, Asia and Africa both had growth, with only Asia being above the world average, while North America incurred a loss. Overall, we can see that growth is far below target for low-income countries, which implies there are long-term structural barriers leading to low productive capacity and weaker institutional presence. In the low-middle-income group, all

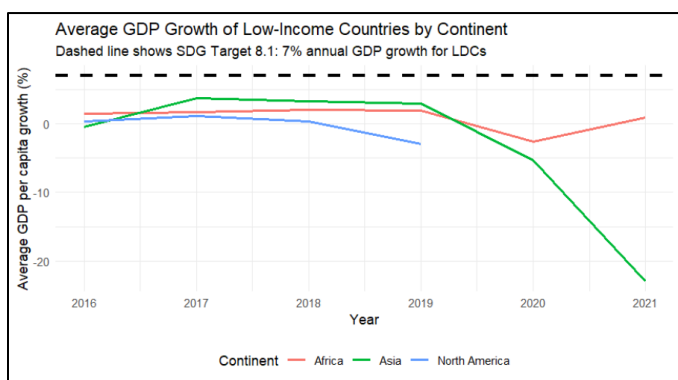


Figure 2.3.1 Year-on-Year Average GDP Growth by Continent (Dashed Line at 7% on Y-Axis)

The only continents with LDCs are North America (1 country), Asia (3 countries) and Africa (26 countries). In all continents, no LDCs are achieving the target of at least 7% GDP growth per annum.

2.4 Discussion

2.4.1 Limitations

Several limitations warrant consideration. GDP per capita does not capture growth quality, inequality, or informal economic activity prevalent in least developed countries. GDP growth rates exhibit substantial year-to-year volatility from commodity prices, weather, and external shocks, with 2020 particularly affected by pandemic recession. Data quality concerns persist in countries with limited statistical capacity, outdated base years, and varying methodologies.

2.5 Conclusion

In conclusion, Target 1 has not been achieved, and is unlikely to meet them even by 2030 based on current trajectories. This is largely due to the detrimental effect of COVID-19 on economic growth worldwide. In no continent have LDCs reached sustained 7% GDP per capita growth, and, due to COVID-19, pre-existing structural gaps widened. However, we believe that this can be solved with policy

intervention. Possible strategies include diversifying away from primary commodities, enhancing domestic resilience to volatility and global shocks, and strengthening institutions that have regulatory power. These strategies could all help LDCs to accelerate their growth. However, significant scaling up is required, implying more investment into said strategies, which would be more difficult for LDCs to achieve.

3.1 Introduction

Youth unemployment and exclusion from education threaten economic development and social cohesion globally. Recognising this challenge, the United Nations established Target 8.6 under Sustainable Development Goal 8, committing member states to achieve a notable reduction in youth not in employment, education or training by 2020.

The NEET indicator captures youth exclusion beyond unemployment, including those withdrawn from job search and unable to access education. This report examines how many countries per continent achieved significant NEET rate reductions between 2015-2019 and 2015-2020, distinguishing policy-driven progress from COVID-19 pandemic impacts.

3.2 Methodology

3.2.1 Research Design

This study evaluates whether the six continents achieved Sustainable Development Goal Target 8.6 which focuses on Youth Employment, specifically aiming to "By 2020, substantially reduce the proportion of youth not in employment, education or training (NEET)" to foster inclusive economic growth, as part of the broader SDG 8: Decent Work and Economic Growth.

The analysis examined NEET rates across 285 countries in six continental regions. Given the strong disruption caused by the COVID-19 pandemic in 2020, two parallel analyses were conducted: 2015 to

2020 for formal SDG target assessment, and 2015 to 2019 to isolate pre-pandemic trends. This dual approach distinguishes policy-driven progress from COVID-19 pandemic effects on youth outcomes.

3.2.2 Threshold for Significant Reduction

A reduction of 3 percentage points or greater was established as the threshold for determining significant decrease in NEET rates. This benchmark was selected for several reasons. First, from a practical policy perspective, a 3-percentage-point reduction represents meaningful progress that affects considerable numbers of young people. Second, this threshold balances ambition with feasibility, as reductions exceeding 5 percentage points are rare within five-year timeframes absent major structural reforms, while thresholds below 2 percentage points may capture measurement variation rather than genuine policy impact. Finally, this absolute threshold provides consistent cross-country comparability regardless of baseline levels, avoiding the complications of relative percentage calculations for countries with very low or very high initial rates.

3.2.3 Analytical Methods

Analysis proceeded in three stages. First, for each country, the change in NEET rate was calculated by subtracting the endpoint year rate (2019 or 2020) from the 2015 baseline rate. Countries demonstrating reductions of 3 percentage points or greater were classified as having achieved significant reduction. Second, countries were grouped by continent to determine the number and proportion of nations within each region meeting the threshold criterion. Third, results were visualized using bar charts displaying the count of countries achieving significant reduction per continent, separately for the 2015-2019 and 2015-2020 comparisons.

3.3 Results

3.3.1 Analysis from 2015 to 2019

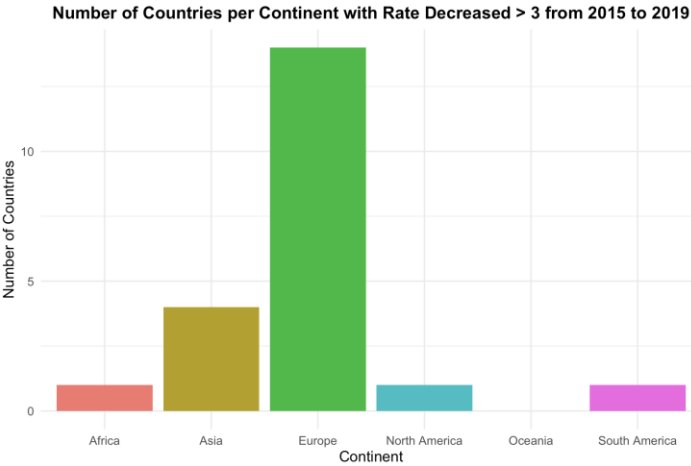


Figure 3.3.1 Number of Countries per Continent with NEET rate decreased $\geq 3\%$ from 2015 to 2019

In the pre-pandemic period (2015-2019), Europe demonstrated the strongest performance with 12 countries achieving the reduction threshold. Asia recorded approximately 4 countries, while Africa, North America, and South America each had approximately 1 country meeting the criterion. Oceania showed no countries achieving significant reductions.

When analysing the period before the pandemic, the results suggest that several continents were making gradual progress in reducing youth NEET rates. Europe stands out, with the highest number of countries registering reductions of more than 3 percentage points between 2015 and 2019. This reflects strong labour market conditions (European Commission, 2023), widespread access to vocational and tertiary education, and effective school-to-work transition systems (European Commission, 2024). Asia also shows moderate improvement, with a noticeable number of countries experiencing declines in NEET rates. This is a result of strong economic performance creating youth jobs paired with expansions in education and vocational training, though progress remains uneven across countries. In contrast, Africa and North America exhibit more limited progress, with relatively few countries achieving substantial reductions before 2020. Africa’s limited reduction in NEET is driven by

structural unemployment (Sibanda and Thebe, 2022), rapid population growth (IMF, 2024; UNESCO, 2025) and political instability. For North America, this is best explained by structural inequality (OECD, 2016) and rising educational costs. Oceania and South America show minimal change, highlighting either persistent structural challenges or more volatile labour market conditions. The instability for Oceania could be caused by small population effects, resource dependence, or uneven youth labour integration. As for South America, economic stagnation and recession destroys youth employment opportunities. The high dropout rates and large informal labour sector also explains the minimal improvement.

3.3.2 Analysis from 2015 to 2020

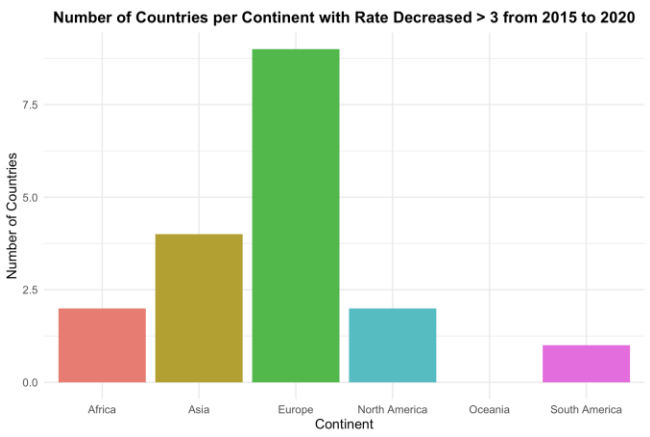


Figure 3.3.2 *Number of Countries per Continent with NEET rate decreased $\geq$ 3% from 2015 to 2020*

When 2020 was included as the endpoint, the distribution shifted notably. Europe declined to approximately 9 countries, suggesting pandemic-related reversals in some nations. Asia maintained approximately 4 countries. Africa and North America each increased to approximately 2 countries, while South America remained at approximately 1 country. Oceania continued to show no countries meeting the threshold.

With 2020 included, the number of countries achieving reductions greater than 3 percentage points drops sharply in some continents. The sharp decline in improving countries between the 2015-2019 and 2015-2020 graphs provides strong evidence that COVID-19 reversed or stalled NEET reductions in most continents. Europe, which showed the strongest progress pre-pandemic, experiences a notable decline in the number of improving countries, indicating that earlier gains were reversed or stagnated during the pandemic year.

Prior to the pandemic, Europe and Asia showed stable progress. The reversal in Europe implies that any progress achieved was fragile rather than structural, as short-term shocks were sufficient to erase multi-year gains. Vulnerable regions like South America saw almost no improvement pre-COVID and the pandemic likely exacerbated existing challenges. Overall, the pandemic appears to have significantly derailed global advancement toward reducing NEET, preventing Target 2 from being meaningfully met by the 2020 deadline.

3.4 Discussion

3.4.1 Limitations

Data quality and collection methodologies varied across countries, potentially affecting comparability. The country-level analysis treats each country as an equal unit regardless of population size, meaning that aggregate progress affecting large populations is weighted equally with changes in smaller nations. The 3 percentage point threshold, while justified, remains somewhat arbitrary and may not capture all meaningful forms of progress. Finally, the five-year timeframe may be insufficient to capture the full impact of structural policy interventions, particularly in regions facing severe development challenges.

3.5 Conclusion

Overall, our findings have shown that most continents did not reduce NEET between 2015 and 2020. Improvements where present were small or reversed — especially in 2020. These patterns suggest that pre-existing progress in reducing youth NEET rates is highly sensitive to external conditions and heavily dependent on stable economic environments. Reducing NEET on a sustained basis therefore requires long-term structural reforms, as well as greater resilience in labour markets and education systems to withstand shocks like the pandemic.

4. Bibliography

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5. Coding

5.1 GitHub Usernames

Yam Yi Pan: yyjessica0730-afk
Rachel Tan: racheltanrq
Bititi Williamson: BititiW
Haris Thrassou: HTICL
Marko Vasiljevic: markoalex2006-star

5.2 Extra CSV File

Original File Name:
“API_LDC_DS2_en_csv_v2_25369.csv”
Renamed file name for convenience:
“World Bank data csv.csv”

5.3 GitHub Link

https://github.com/BititiW/IDS_Group_Assignment

6. Appendices

Appendix A: Heat maps showing annual GDP growth for all continents from 2015 to 2019

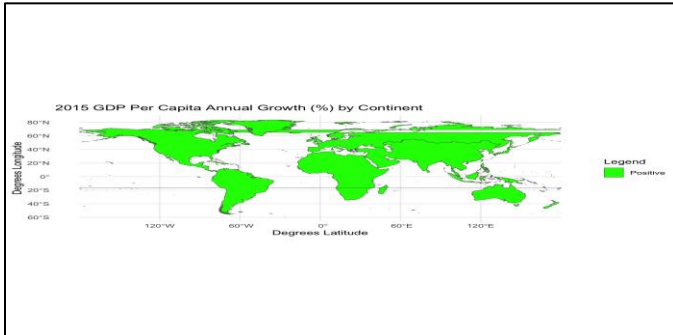


Figure A1

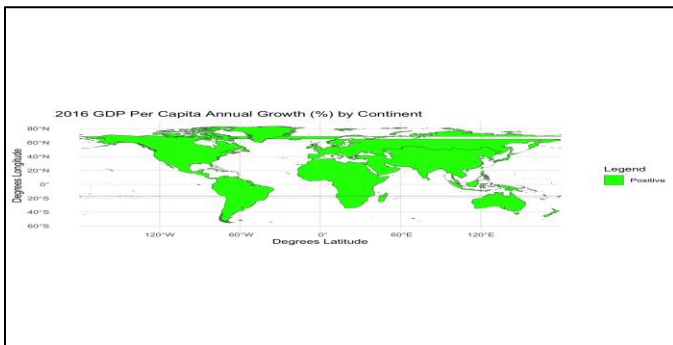


Figure A2

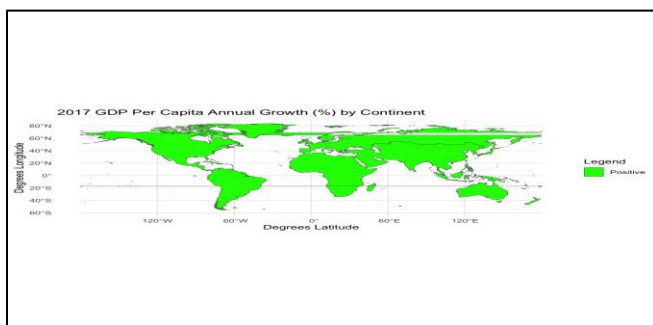


Figure A3

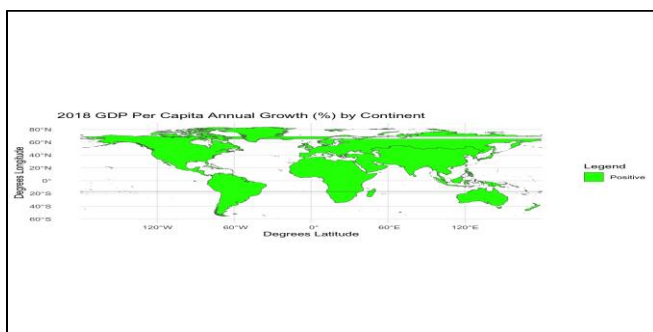


Figure A4

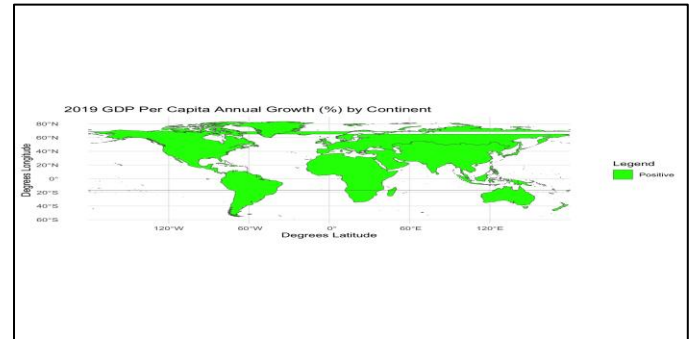


Figure A5